MIL-L-45922(WC)

8 March 1971

SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

LAUNCHER, GRENADE, 40MM: XM129

1. SCOPE

1.1 This specification covers one type of 28 volts direct current, 40-millimeter, electric motor driven, link-belt fed, automatic grenade launcher, XM129.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	-	Preservation, Methods of.
MIL-C-372	-	Cleaning Compound, Solvent (For Bore of Small Arms and Automatic Aircraft Weapons).
MIL-G-10944	-	Gages, Dimensional Control.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

FSC 1010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army Weapons Command		
A11696982	-	XM129 Grenade Launcher Assembly for XM28 Subsystem.
C11697135	-	General Requirements for Assembly Lubrication of 40mm Grenade Launcher.

PUBLICATIONS

U.S. Army Weapons Command		
IEL11696982	-	List of Inspection Equipment Numbers.
PDS11696982	-	Packaging Data Sheet.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1.1 First article sample. Requirement for submission of the first article shall be as specified in the contract (see 6.1). One of the packs shall be considered the pilot pack (see 5.1).

3.2 Materials and construction. Launchers and parts shall conform to the material and construction requirements specified herein and on drawings referenced on Drawing A11696982 and in accordance with the applicable materials and construction provisions of MIL-W-13855.

3.3 Design. Launcher and parts shall conform to the design specified herein and on drawings referenced on Drawing A11696982 and in accordance with the applicable design provisions of MIL-W-13855.

3.3.1 Slide assembly. The slide actuator arm shall be pinned to the cam arm shaft, and when assembled to the feed support with the bearings in place, shall swing freely. The cam follower gear shall mesh with the slide actuator gear without binding and, with the gears meshed, the arms shall move freely through their full range of travel.

3.3.1.1 Arm and gear assembly. The feed cam roller shall be retained by the spring pin and shall rotate freely in the cam follower arm.

3.3.2 Feed slide assembly. The cartridge feed pawl shall pivot with no restriction and, when depressed, shall return to its extended position and be held by the helical compression spring and plunger. The feed slide shall move smoothly and freely through the range of its travel in the feed slide slot in the receiver.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T04:15Z
Check the source to verify that this is the current version before use.

3.3.3 Plunger assembly. The cartridge stop plunger shall slide freely within the spring retainer, when assembled into the receiver with the spring and the spring pin. When depressed, the plunger shall return without binding and be held in its forward position by the compression spring.

3.3.4 Round stop assembly. The cartridge stop shall move freely within its guide hole in the receiver. When engaged with the cartridge stop lever, it shall be held in the aft position by the plunger spring. When the plunger is depressed, the cartridge stop shall move forward. When the plunger is released, the cartridge stop shall return and remain in the aft position.

3.3.5 Round positioners. The round positioners shall slide freely in the reciever. When depressed, the positioners shall return to their full extended position without binding and be held there by the compression springs.

3.3.6 Breech insert assembly. The breech insert and firing pin spring, held in place by a spring pin, shall maintain the firing pin in its extreme aft position. When depressed forward, the firing pin shall move freely in the breech insert and, upon release, shall return and remain in the aft position. The firing pin cushion shall be retained in position by its own resilience.

3.3.7 Striker. The striker shall fit into the guide slot in the receiver without binding and shall move forward and aft through its entire range of travel without binding.

3.3.8 Sear assembly. The sear shall be assembled to the sear carrier by the spring pin, shall pivot of its own weight, and swing freely within the slot of the sear carrier. The sear assembly shall be engaged with the striker by the spring guide, when assembled in the guide slot. The sear carrier shall move freely in the seared and unseared conditions of motion.

3.3.9 Sear release. The sear release shall be assembled to the receiver by the spring pin and shall pivot freely of its own weight. When engaged with the sear, the compression spring shall cause the sear release to maintain a downward force on the sear, but such force shall not prevent the forward and aft travel of the sear. When the forward end of the release lever is depressed by cam action, it shall pivot, compress the sear release spring, and lift the sear away from the striker, allowing the striker to hit the firing pin.

3.3.10 Hammer block. The hammer block, when assembled to the spring block, shall move freely in the guide hole in the drive support casting and the spring block shall move freely on the spring guide rods.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:15Z
Check the source to verify that this is the current version before use.

3.3.11 Barrel lock. The barrel lock shall pivot of its own weight and move freely within the slot of the bracket. When the bracket is fastened tightly to the receiver, the lock shall move without binding into and out of the barrel lock clearance hole in the receiver. With the barrel assembled and in its extreme aft position in the receiver, the lock shall engage to the slot in the barrel.

3.3.12 Yoke guide. The yoke guide shall be held securely in the receiver by the inner bearing races and shall be prevented from rotating by the protrusion of the guide rails through the longitudinal slot of the receiver.

3.3.13 Barrel assembly. The barrel assembly shall slide freely in the yoke guide and the barrel guide assembly after installation into the receiver.

3.3.13.1 Barrel. The barrel shall be free of cracks and seams. The bore and chamber shall be free of foreign matter, corrosion, burs, pits, bulges and deformation. The chrome plating shall be free of nodules, flaking, stripping and anode burns, and evidence of etched base steel. No mechanical methods for removal of chrome plating are permitted in the barrel bore and chamber. Entrance to the chamber shall be free of burs and sharp edges.

3.3.13.2 Yoke. The yoke shall be securely fastened to the barrel by the yoke key, retaining nut, and retaining pins. The barrel guide rollers shall rotate freely upon the yoke shaft and be retained by a pinned ring type retainer. The cam roller shall rotate freely after assembly to the yoke and roller retainer.

3.3.14 Front plate assembly. The front plate assembly shall fit secruely into the forward end of the barrel cam. The bearing cup of the forward bearing shall be press-fitted into the front plate.

3.3.15 Support assembly. The dust cover shall fit over the outside diameter of the bearing cone so that the feathered edge of the cover shall lie flat against the beveled surface of the front plate. The front support assembly shall be assembled against the dust cover and retained by the barrel cam retaining nut. The retaining nut shall be locked to the receiver by a headless set screw. The set screw shall be seated, below the outer surface of the retainer, permitting the support assembly to rotate without binding in the retainer. The support rollers shall be assembled to the front support by the retaining rings and shall rotate freely.

3.3.16 Drive assembly. The drive support and gear bracket shall be alined to each other by dowel pins and securely fastened together with machine screws. The drive shaft shall be keyed to the drive gear and pinned to the headed straight pin which retains the gear on the shaft. The gears shall mesh without binding and the drive shaft shall rotate freely by hand through 360°.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:15Z
Check the source to verify that this is the current version before use.

3.3.17 Headspace. The headspace in the assembled launcher shall be not less than 2.3872 inches nor more than 2.4376 inches when measured to the 1.616 inch diameter datum on the first shoulder of the chamber.

3.3.18 Firing pin indent. The firing pin indent shall be not less than .016 inch nor more than .025 inch and shall not be off center more than one-half the diameter of the firing pin point.

3.3.19 Interchangeability. Unless otherwise specified on the Drawings, all parts shall be interchangeable.

3.3.20 Fasteners. All fasteners shall be torqued, lock wired and secured as specified on the applicable drawings.

3.4 Performance.

3.4.1 Power input. The launcher shall meet the requirements specified herein with an input voltage of 28 volts direct current (vdc) ± 0.5 vdc.

3.4.2 Rate of fire. The launcher shall be capable of a rate of fire of 420 shots per minute plus or minus 20 shots per minute.

3.4.3 Dry cycling. The launcher shall be capable of continuous dry cycling (operating without ammunition) for a minimum of one minute.

3.4.4 Function firing. The launcher shall be capable of firing 25 rounds in one continuous burst without malfunction.

3.4.5 Accuracy. At a range of 1,000 inches, 25 rounds fired in one continuous burst shall group within a 9.0 inch diameter circle.

3.4.6 Endurance. The launcher shall be capable of firing 5,000 rounds in accordance with the requirements of this specification with no malfunctions or unserviceable parts.

3.4.7 Ammunition and links. The launcher shall meet the requirements specified herein using Government standard 40mm, M385 practice cartridges and Government standard 40mm, M16 links.

3.5 Marking. The launcher shall be marked in accordance with the applicable drawing and MIL-W-13855.

3.6 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:15Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article sample inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.4 First article sample inspection. Inspection of the first article sample (see 3.1) and pilot pack (see 5.1) shall be selected from early production and submitted for testing in accordance with the contract requirements. The sample shall be a representative of production processes to be used during quantity production. The first article sample shall be subjected to all inspection and tests specified herein, and such other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. The formation, size, and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, and contractor's delivery schedule, and within the limitations of MIL-W-13855. Launchers shall be assembled from lots of component parts that have met all inspection requirements specified herein. Inspection lot size for the firing pin indent test, the endurance test and the interchangeability test shall be as specified in 4.5.3.2, 4.5.3.4, 4.5.3.5 respectively.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be as specified in the contract (see 6.1). The contractor's examination of these parts shall be prior to and during their assembly into the end items or prior to submission for acceptance as repair parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

4.5.2.2 Launchers. Final examination of each launcher shall be performed after completion of all testing and prior to preservation and packaging. Each launcher shall be visually examined for proper cleaning, presence of protective coating, completeness of manufacture, workmanship and shall meet the requirements specified in Table I.

4.5.2.2.1 Visually and manually examine the following to determine conformance to applicable requirements:

TABLE I

	Ref Para
Slide assembly	3.3.1, 3.3.1.1
Feed slide assembly	3.3.2
Plunger assembly	3.3.3
Round stop assembly	3.3.4
Round positioners	3.3.5
Breech insert assembly	3.3.6
Striker	3.3.7
Sear assembly	3.3.8
Sear release	3.3.9
Hammer block	3.3.10
Barrel lock	3.3.11
Yoke guide	3.3.12
Barrel assembly	3.3.13, 3.3.13.1, 3.3.13.2
Front plate assembly	3.3.14
Support assembly	3.3.15
Drive assembly	3.3.16
Fasteners	3.3.20

4.5.2.2.2 All markings shall be visually examined for clarity, legibility and compliance with the applicable drawing. The identification plate shall be examined for secure assembly.

4.5.2.2.3 Inspection approval stamp. Launchers which have passed all examinations after sucessful completion of testing shall be stamped with the Department of Defense complete inspection approval stamp as specified on the applicable drawing. The Government representative will control the stamps used for such stampings to assure successful completion of all required examinations and tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

4.5.2.3 Packaging. Examination of packaging of launchers shall be performed in accordance with the classification of defects and acceptable quality levels (AQL's) specified in 4.5.2.3.1. Sample size shall be in accordance with MIL-STD-105. The following provisions shall apply:

a. The AQL's are specified as percent defective.
b. The AQL listed for each defect shall be applied to the individual defect, not to a group of defects.
c. Examination for packaging defects specified in 4.5.2.3.1 shall apply to each sample of launchers or packages as applicable.

4.5.2.3.1 Classification of defects for packaging. Unless otherwise specified, the packaging requirements are specified on Packaging Data Sheet P11696982.

Categories	Defect	AQL
Critical:	None defined.	
Major:		
101	Illegible or incorrect marking.	1.0
102	Improper location of marking.	1.0
103	Improper level of packaging and packing (see procurement documents).	1.0
104	Inadequate cleaning and drying.	1.5
105	Improper assembly of package (position of launchers in support, and improper positioning of cushioning material).	1.5
106	Improper closure of interior package.	1.5
107	Improper closure of shipping containers.	1.5
Minor:		
201	Workmanship	

4.5.2.3.2 Packaging of repair parts. Examination for packaging of repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1).

4.5.3 Testing.

4.5.3.1 Headspace testing. Each launcher shall be tested using the test method specified in 4.6.1 to determine compliance with the requirements of 3.3.17. Launchers failing to meet the requirement shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

4.5.3.2 Firing pin indent testing. A sample of five launchers selected by the Government representative from each month's production shall be tested using the test method specified in 4.6.2 to determine compliance with the requirements of 3.3.18. Failure of any launcher to meet the requirements shall be cause for rejection of the entire month's production.

4.5.3.3 Rate of fire, dry cycling, function firing and accuracy testing. Each launcher shall be tested using the test method specified in 4.6.3 to determine compliance with the requirements of 3.4.2, 3.4.3, 3.4.4 and 3.4.5. Launchers failing to meet the requirements shall be rejected.

4.5.3.4 Endurance testing.

4.5.3.4.1 Lot size. The initial lot of launchers shall consist of the first month's production. Thereafter, unless otherwise specified (see 6.1) each endurance test lot shall consist of 100 launchers or a month's production, whichever is smaller.

4.5.3.4.2 One launcher selected by the Government representative from each inspection lot shall be tested by the contractor using the test method specified in 4.6.4 to determine compliance with the requirements of 3.4.6. If the test requirements are not met, acceptance of the launchers shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.
c. Examine launchers, partially assembled launchers, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Upon instruction from USAWECOM, Attn: AMSWE-QA, resume testing of the launcher, or retest an additional launcher from the same lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

4.5.3.5 Interchangeability testing.

4.5.3.5.1 In plant. The supplier shall test a sample of five launchers, randomly selected by the Government representative, from each month's production for interchangeability using the test method specified in 4.6.5.1. Launchers taken for interchangeability testing shall have been found satisfactory in all other examinations and tests, including function firing and accuracy, headspace, barrel lock engagement and firing pin indent. The launchers shall be examined and tested for headspace, barrel lock engagement, firing pin indent and function firing and accuracy after interchange of parts. Failure of the interchangeability test shall be cause for rejection of the entire lot. Prior to submission as a reconditioned lot, supplies correction shall be effected on all launchers of the lot and on all launchers and component parts not yet accepted. Sample size and test method for reconditioned lots shall be the same as for the original lot.

4.5.3.5.2 Interplant. When launchers are manufactured concurrently by more than one supplier, each supplier shall forward, to a testing agency designated by the procuring agency (see 6.1), five launchers monthly for the interplant interchangeability test specified in 4.6.5.2. Launchers submitted for interchangeability testing shall be examined and tested for function firing, accuracy, headspace, barrel lock engagement and firing pin indent. The launchers shall be examined and tested for headspace, barrel lock engagement, firing pin indent and function firing and accuracy after interchangeability tests. The supplier will be informed of any failure of the launchers to meet prescribed requirements. Upon completion of inspection by the testing agency, samples shall be packaged level C and will be returned to the supplier for repackaging in accordance with procurement documents at the supplier's expense.

4.5.3.6 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.1). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.3.7 Packaging testing.

4.5.3.7.1 Launchers.

4.5.3.7.1.1 The supplier shall furnish the Government representative with cartification that the packaging materials conform to the applicable packaging data sheets and specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

4.5.3.7.1.2 Determination of cleanliness. The supplier shall test items from each inspection lot for determination of cleanliness as specified below. Sampling shall be in accordance with MIL-P-116.

a. The applicable surfaces of each sample unit shall be subjected to the determination of cleanliness test specified in MIL-P-116, except that the wipe test shall not be applicable to parts with anodized protective coatings.

4.5.3.7.2 Containers.

4.5.3.7.2.1 Quick leak. The contractor shall test containers from each inspection lot for quick leak using the test methods and sampling as specified in MIL-P-116.

4.5.4 Inspection equipment. Unless otherwise specified in the procurement documents (see 6.2), responsibilities for acquisition, maintenance, and disposition of measuring and testing equipment prescribed on the Index of Inspection Equipment Lists IEL11696982 and for all other inspection equipment required to perform inspection required by applicable specifications, shall be in accordance with MIL-G-10944 and MIL-I-45607.

4.6 Test methods.

4.6.1 Headspace test. This test shall be performed after completion of final assembly and prior to function firing test using inspection equipment listed on IEL11696982. With the plug inserted in the barrel chamber and the barrel in battery position, the minimum headspace gage shall enter between the plug and the breech insert face, and the maximum headspace gage shall not enter between the plug and the breech insert face.

4.6.2 Firing pin indent test. Launchers shall be tested using inspection equipment listed on IEL11696982. The length of each compression cylinder shall be measured prior to indent. The launcher shall be held with the copper compression cylinder holding fixture with the test cylinder inserted in the barrel chamber. The barrel cam assembly shall be manually rotated in the direction specified on the decal to move the barrel into the battery position releasing the striker and indent the copper cylinder. The barrel cam assembly shall be further rotated in the same direction to move the barrel out of battery position and the holding fixture shall then be removed from the barrel. The depth of the indent in the copper cylinder shall be determined by measuring the distance for the original surface to the bottom of the firing pin impression. Depth measurements may be performed using standard depth indicator or other

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

standard measuring equipment. The firing pin shall be measured on three copper compression cylinders for each sample launcher. The results shall be recorded and the average depth of the three indents shall be calculated to determine whether the requirement has been met. Each indent shall be visually examined to determine whether the concentricity requirement has been met.

4.6.3 Rate of fire, dry cycling, function firing and accuracy test. Launchers shall be tested at a range of 1,000 inches using inspection equipment listed on IEL11696982.

4.6.3.1 Manually rotate the barrel cam in the direction of normal operation and visually examine moving parts to assure proper operation.

4.6.3.2 Cycle each launcher without ammunition for one minute with 28.0 vdc ± 0.5 vdc applied to the drive motor.

4.6.3.3 One belt of 25 rounds of ammunition and links conforming to 3.4.7 shall be loaded.

4.6.3.4 The 25 rounds shall be fired in one continuous burst. Rate of fire shall be checked and recorded during 10 rounds of the 25 round burst.

4.6.3.5 The target shall be checked to determine compliance with accuracy performance criteria.

4.6.4 Endurance test. Launchers shall be tested using inspection equipment listed on IEL11696982.

4.6.4.1 Voltage applied to the launcher drive motor shall be 28.0 vdc ± 0.5 vdc.

4.6.4.2 Ammunition and links shall be in accordance with 3.4.7.

4.6.4.3 A total of 5,000 rounds shall be fired from each endurance test launcher.

4.6.4.4 The ammunition shall be linked together in belts of 25 rounds each. Each 25 rounds will be fired in one continuous burst. The rate of fire shall be measured and recorded during 10 rounds of the first belt fired following each field strip operation.

Source: https://assist.dla.mil -- Downloaded: 20260-22T04:15Z
Check the source to verify that this is the current version before use.

4.6.4.5 The launcher shall be visually examined after firing 150, 300, and 450 rounds. Areas that have carbon and dirt accumulations shall be wiped or brushed clean. The rifles tube and chamber portions of the barrel shall be thoroughly cleaned to remove all traces of copper deposit using phosphor-bronze bristle brush and rifle bore cleaner per MIL-C-372.

4.6.4.6 At the end of each 600 rounds of firing, the launcher shall be field stripped, cleaned, inspected, and lubricated in accordance with procedure below. Commercial parts (pins, screws, etc) that become damaged during field strip operations may be replaced and shall not be counted as a malfunction of the weapon.

a. Disassemble the feed actuator key, drive assembly, firing mechanism, sear carrier assembly, top round positioners, feed actuator assembly, barrel cam retainer and front support assembly, dust cover, barrel guide assembly, front plate and bearings, barrel assembly, barrel cam assembly, yoke guide, lock and feed cam assembly, barrel lock assembly, breech insert, firing pin and spring and lower round positioners.

b. Thoroughly clean all parts removed and manually operate moving parts to assure proper operation.

c. Visually examine to determine that parts are neither broken or worn to the extent that they are unserviceable. An unserviceable part is one that causes malfunctions.

d. Reassemble the launcher, applying lubricant in accordance with Drawing C11697135.

e. After assembly, manually operate the launcher to assure proper operation of all parts.

4.6.4.7 A complete record shall be kept for each endurance test, showing each malfunction and all parts replaced, including the number of the round at which the part replacement occurred. Malfunctions attributable to ammunition or links shall not be counted against the launcher being tested. However, such malfunctions shall be recorded. Components damaged as a result of ammunition or link stoppages may be replaced and shall not be counted as a malfunction of the weapon.

4.6.4.8 Endurance tested launchers and parts shall be disposed of as specified in the contract (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

rakes - With the guides loaded and positioned as 3.3.1/ Power-off br

4.6.5 Interchangeability test.

4.6.5.1 Inplant. Launchers shall be tested for interchange of parts by disassembling and then reassembling the parts using the parts and prearranged system specified below. Interchange of parts shall be accomplished by dividing the parts of each launcher into 5 groups of non-mating parts distributing the groups into 5 different trays until each tray contains parts for a complete launcher. Groups of parts from the first launcher shall be taken in order and placed in trays 1, 2, 3, 4 and 5; groups of parts from the second launcher shall be taken in order and placed in trays 2, 3, 4, 5 and 1; groups of parts from the third launcher shall be taken in order and placed in trays 3, 4, 5, 1 and 2; groups of parts from the fourth launcher shall be taken in order and placed in trays 4, 5, 1, 2 and 3; groups of parts from the fifth launcher shall be taken in order and placed in trays 5, 1, 2, 3 and 4. Commercial parts such as screws, nuts, washers and pins shall be placed in the same tray or area as their mating or associate part. Any commercial part rendered unserviceable by disassembling shall be replaced without penalty to the interchangeability test. The launchers shall be reassembled using only those parts which are in the same tray or area.

4.6.5.2 Interplant. Launchers to be subjected to the interplant interchangeability test shall be manually functioned to assure proper operation before parts are diassembled from the launcher. Launchers shall be interchanged in accordance with system specified in 4.6.5.1 except that when disassembling, every other launcher used shall be one produced by a different supplier. Parts shall be identified with their respective manufacture throughout the test. Before launchers are returned to the suppliers the original parts shall be reassembled to their respective launchers and the launchers given a hand functioning test to assure proper operation.

GROUPS OF NONMATING PARTS

Group I

11697136	Roller	11607091	Barrel
11697028	Slide, Feed	11697058	Retainer, Barrel Guide Roller
11697144	Hammer, Block	11696993-1	Drum, Assy
11696992(2)	Bearing, Ball	11697078	Lever, Cartridge Stop
11696991	Bearing, Ball	11697103	Insert, Breech
11696984	Shaft, Drive		
11697080(4)	Spring		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

Group II

11696998	Support, Feed
11697029	Pawl, Feed
11697106	Gear, Interim
11697101	Sahft, Retainer Pin
11697145(2)	Rod, Spring Guide
11697081(4)	Positioner, Cartridge
11697035	Carrier, Sear
11697062	Guide, Barrel
11697052	Guide, Barrel
11697053	Nut, Yoke Retaining
11697057	Roller, Barrel Guide
11697127-1	Barrel, Lock Assy
11697025	Cam, Feed
11697071	Stop, Round
11697041	Cushion, Firing Pin

Group III

MS20392-2057	Pin
11696999	Lever, Feed
11697001	Sahft, Cam Arm
11697030	Pin Pawl Feed
11697104	Gear, Drive
11697146(2)	Spring
11697040	Striker
11696983	Spring
11697063	Insert, Barrel Guide
11697056	Yoke, Barrel Roller Shaft
11697124	Roller
11697067	Bearing, Cone Rear
11697125	Release, Sear
11697079	Pin, Cartridge Stop Lever
11697112	Spring, Helical Compression
11697082	Retainer, Cartridge Positioner Spring

Group IV

11697139(2)	Plunger
11697000	Gear, Feed
11696986	Key, Feed Actuator
11697032	Spring
11697148	Support Assy, Drive
11697037	Guide, Carrier Spring
11697070	Retainer, Barrel Cam
11697084(2)	Roller, Front Support
11697069-1	Bearing, Cone Front
11697059	Retainer, Roller Barrel Cam
11697094	Pin, Yoke Retaining
11697050	Guide, Yoke
11697024	Cam (See note)
11697074-1	Plunger Assy
11697042	Pin, Firing

Note: Do not remove pin 11097026 from cam 11697024.

Group V

11697095	Cover, Dust
11697002-1	Arm and Gear Assy Follower
11697031	Plunger
11697107	Bracket, Gear
11697147	Block, Spring
11696990	Spring, Helical Compression
11697036	Sear
11697097	Support, Front
11697064-1	Plate, Assy Front
11697060	Roller, Barrel Cam
11697068	Receiver, Machine

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

4.6.5.3 Upon successfilly completing the interchangeability test, the launchers-parts, shall be submitted for final inspection and acceptance as part of the month's production.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack, consisting of a complete and packed unit and intermediate packages, as applicable, shall be packaged in accordance with Packaging Data Sheet P11696982 for the level of protection specified in the contract (see 6.1) and packed level C, shall be forwarded as specified in 3.1.

5.2 Preservation, packaging, packing and marking. Launchers shall be preserved, packaged, packed and marked in accordance with the requirements of Packaging Data Sheet P11696982 for the level of protection specified (see 6.1).

5.3 Repair parts. Repair parts shall be prepared for delivery in accordance with the applicable packaging data sheets.

6. NOTES

6.1 Ordering data.

a. Title, number, and date of this specification.
b. List of drawings and specifications pertinent to the launcher, showing applicable revision dates.
c. Number of samples required and shipping instructions for first article sample and pilot pack (see 3.1 and 5.1).
d. Lot size for endurance test if different (see 4.5.3.4.1).
e. Agency to be notified in case of endurance test failure (see 4.5.3.4.2).
f. Disposition of endurance tested launchers (see 4.5.3.4.2).
g. Shipping instruction for launchers for interchangeability test (see 4.5.3.5.2).
h. List of inspection equipment, responsibilities for acquisition, maintenance, and disposition of, if other than as specified (see 4.5.4).
i. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
j. Packaging instructions for repair parts (see 5.3).
k. Criteria for examination and testing of component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.6).
l. Responsibilities for furnishing ammunition and links.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

6.2 Test firing facilities and operating procedures shall be designed by the contractor in conformance with local, state, and federal regulations and suitable for carrying out prescribed firing tests with safety of operating and visiting personnel. Copies of these contractor designs shall be forwarded to the contracting officer. Government facilities may be viewed upon application to the contracting officer.

6.3 When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

6.4 To avoid delay in test firing, the Government representative should maintain a minimum of two months' supply of ammunition as determined by anticipated firing requirements.

6.5 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.6 Supersession data. This specification includes the requirements of USAWECOMPD-10 dated 12 July 1968.

6.7 Unless otherwise specified (see 6.1 h), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Summary List, DD Form 1660.

Custodian:
Army - WC

Preparing activity:
Army - WC

Project number:
1010-A058

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-L-45922(WC) Launcher, Grenade, 40MM: XM123

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
HEADQUARTERS
U. S. ARMY WEAPONS COMMAND

ROCK ISLAND, ILLINOIS 61201

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Weapons Command
ATTN: AMSWL-REE-S
Rock Island, Illinois 61201

FOLD

To detach this form, cut along this line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:15Z
Check the source to verify that this is the current version before use.